

AES128 FPGA Core Project Overview Documentation

This document provides a detailed overview of the AES128 FPGA Core project implemented using Verilog HDL and validated on FPGA hardware using Xilinx Vivado and Integrated Logic Analyzer (ILA). The project was developed as a collaborative FPGA-based cryptographic hardware implementation focusing on AES encryption, AES decryption, FPGA debugging, datapath visualization, and hardware verification.

1. Introduction

The Advanced Encryption Standard (AES) is one of the most widely used symmetric encryption algorithms in modern digital systems. AES is extensively utilized in secure communication systems, embedded systems, banking applications, cloud infrastructure, wireless communication, military systems, and hardware security implementations. The objective of this project was to implement a complete AES-128 encryption and decryption datapath using Verilog HDL and verify its operation on FPGA hardware. Unlike many basic AES mini projects which focus only on software simulation or partial encryption stages, this project focused on complete hardware-level verification and debugging using FPGA tools. The implementation includes both encryption and decryption modules along with intermediate AES transformations such as SubBytes, ShiftRows, MixColumns, inverse transformations, S-Box lookup tables, inverse S-Box lookup tables, and Vivado ILA-based signal visualization.

2. Project Objectives

The primary objectives of the project were:

- To implement AES-128 encryption using Verilog HDL.
- To implement AES-128 decryption using inverse AES transformations.
- To develop FPGA-compatible RTL modules for all AES stages.
- To integrate encryption and decryption into a complete AES hardware system.
- To verify functionality through simulation and FPGA implementation.
- To perform hardware-level debugging using Vivado ILA.
- To visualize intermediate AES transformation stages in real time.
- To validate successful plaintext recovery after decryption.

3. Why AES?

AES (Advanced Encryption Standard) is a symmetric block cipher standardized by NIST and widely accepted as a secure cryptographic algorithm for protecting digital data. AES-128 specifically operates on 128-bit plaintext blocks using a 128-bit key. AES was selected for this project because:

- It is one of the most important modern cryptographic algorithms.
- It provides strong security and resistance against cryptographic attacks.
- It contains mathematically rich hardware transformations suitable for FPGA implementation.
- It allows exploration of finite field arithmetic and hardware optimization techniques.
- It is highly relevant in modern VLSI and embedded system applications.

4. Why FPGA Implementation?

FPGA implementation offers significant advantages for cryptographic systems:

- Hardware parallelism improves computational efficiency.
- FPGA-based cryptography provides higher throughput compared to software implementations.
- Real-time signal monitoring is possible using hardware debugging tools.
- FPGA implementation helps in understanding low-level hardware datapath operations.
- Hardware accelerators are widely used in secure embedded systems and communication devices.

The project was implemented using Verilog HDL and synthesized using Xilinx Vivado targeting FPGA hardware.

5. System Architecture

The AES128 FPGA Core consists of three major sections: 1. Encryption Module 2. Decryption Module 3. FPGA Debugging and Verification Infrastructure Encryption Flow: Plaintext → AddRoundKey → SubBytes → ShiftRows → MixColumns → AddRoundKey → Ciphertext Decryption Flow: Ciphertext → AddRoundKey → InvMixColumns → InvShiftRows → InvSubBytes → AddRoundKey → Recovered Plaintext The complete datapath was integrated under a top-level system module and verified through both simulation and FPGA execution.

6. FPGA Debugging using Vivado ILA

One of the major highlights of the project was the use of Vivado Integrated Logic Analyzer (ILA) for hardware debugging and datapath visualization. The following signals were monitored in hardware: • Plaintext • SubBytes output • ShiftRows output • MixColumns output • Ciphertext • InvMixColumns output • InvShiftRows output • InvSubBytes output • Recovered plaintext ILA-based debugging allowed real-time verification of internal AES stages directly on FPGA hardware.

7. Simulation and Verification

The system was extensively verified using Verilog testbenches before FPGA implementation. The following verifications were performed: • Encryption correctness verification • Decryption correctness verification • Intermediate transformation validation • End-to-end plaintext recovery • FPGA hardware validation The plaintext “HELLO WORLD” was successfully encrypted into ciphertext and later decrypted back to the original plaintext, confirming correct functionality of the complete AES datapath.

8. Project Challenges

Several technical challenges were encountered during development: • S-Box and inverse S-Box memory initialization • Correct inverse AES transformation sequencing • FPGA synthesis issues • Vivado ILA probe mismatches • Datapath integration challenges • Hardware-level debugging • Signal stabilization and verification These issues were resolved through iterative debugging, simulation analysis, FPGA verification, and datapath correction.

9. Technologies Used

The following technologies and tools were used in the project: • Verilog HDL • Xilinx Vivado • Vivado Integrated Logic Analyzer (ILA) • FPGA Hardware Platform • AES-128 Cryptographic Algorithm • RTL Design Methodology • Hardware Verification Techniques

10. Final Outcome

The project successfully demonstrated a complete FPGA-based AES128 encryption and decryption system with real-time hardware verification. The final system achieved: • Functional AES encryption • Functional AES decryption • Successful plaintext recovery • FPGA implementation and execution • Real-time hardware debugging using ILA • Complete datapath verification The project evolved beyond a conventional academic mini project and

became a complete FPGA-based cryptographic hardware implementation demonstrating both theoretical and practical aspects of hardware security systems.

11. Development Team

Team Member	Contribution
Kaushal Naik	MixColumns and InvMixColumns implementation
Dipiksha Parab	ShiftRows and InvShiftRows implementation
Vedam Panjekar	SubBytes, S-Box and lookup-table implementation
Jaydev Rajpurohit	InvSubBytes, inverse S-Box, key handling and testing support
Jatin (Team Lead)	System integration, debugging, FPGA implementation, Vivado ILA integration, verification

This document serves as the high-level overview of the AES128 FPGA Core project. Additional documentation files provide detailed explanations of the AES algorithm, RTL implementation, FPGA flow, simulation methodology, hardware verification, and Vivado ILA debugging process.